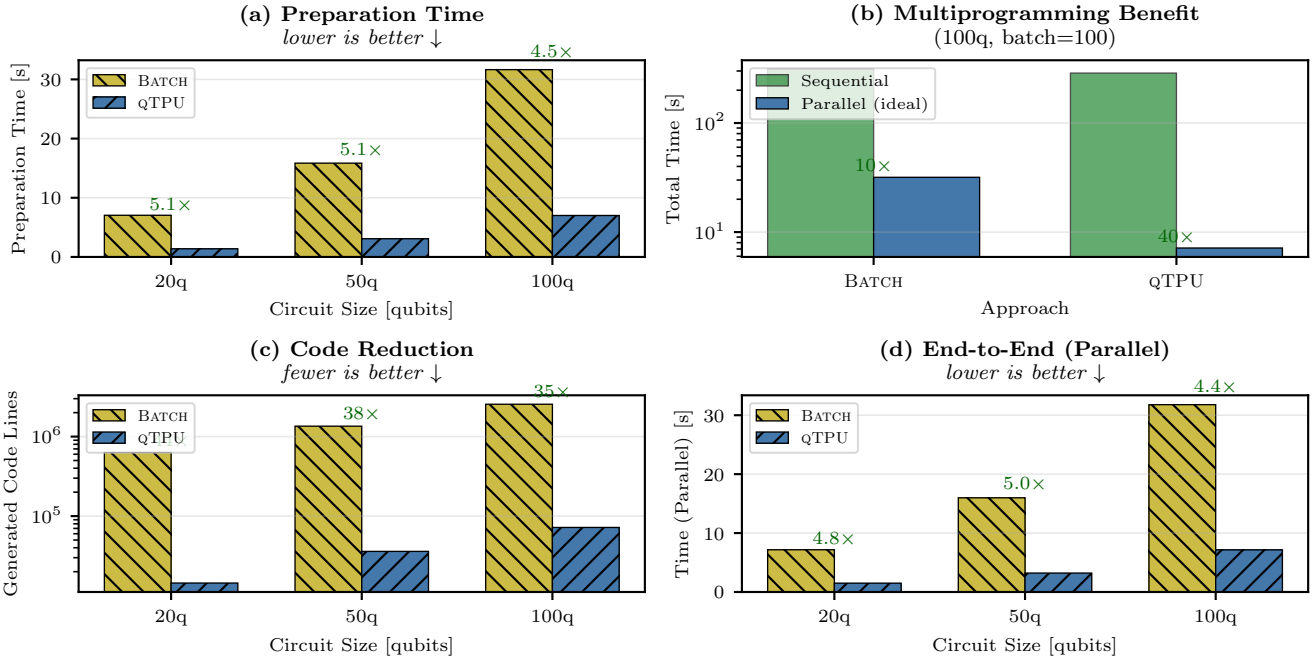
(a) Preparation Time
lower is better ↓
Preparation Time [s]
Batch
qTPU
5.1×
5.1×
4.5×
20q
50q
100q
Circuit Size [qubits]
(b) Multiprogramming Benefit
(100q, batch=100)
Total Time [s]
Sequential
Parallel (ideal)
10×
40×
Batch
qTPU
Approach
(c) Code Reduction
fewer is better ↓
Generated Code Lines
Batch
qTPU
38×
35×
20q
50q
100q
Circuit Size [qubits]
(d) End-to-End (Parallel)
lower is better ↓
Time (Parallel) [s]
Batch
qTPU
4.8×
5.0×
4.4×
20q
50q
100q
Circuit Size [qubits]